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DetectWear: AI Based Human-Centric Wearable Number Detection In CCTV Streams

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BONAFIDE CERTIFICATE

This is to certify that this project/internship work entitled **DetectWear: AI Based Human-Centric Wearable Number Detection In CCTV Streams** is a bonafide record of the technical work carried out by Ms. **Parthi Parinita Karuna Malisetty** (College Reg. No: **23R01A66X2**, SDSC SHAR Trainee ID: **PRJ2026089**) pursuing **IV** year from **CMR Institute of Technology, Kandlakoya, Hyderabad** under my supervision from **04/05/2026** to **11/06/2026** at **Range Operations (RO)** in **Satish Dhawan Space Centre SHAR, Sriharikota**, during the year **2026 - 2027**.

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DECLARATION

I, **Parthi Parinita Karuna Malisetty**, student of **CMR Institute of Technology, Kandlakoya, Medchal, Hyderabad, Telangana**, hereby declare that the project entitled "**DetectWear: AI Based Human-Centric Wearable Number Detection In CCTV Streams**" is an authentic record of the industrial training and technical work carried out by me at **Satish Dhawan Space Centre (SDSC) SHAR, Indian Space Research Organization (ISRO), Sriharikota**.

The matter embodied in this report is original and has not been submitted for the award of any other degree or diploma elsewhere. All technical contributions to the architecture and development of the software reflect my original engineering work performed under the guidance of the **Range Operations (RO)** department, **Satish Dhawan Space Centre (SDSC) SHAR, ISRO, Sriharikota**.

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ABSTRACT

This project presents the design and implementation of **DetectWear**, an intelligent deep learning-based industrial surveillance system developed for real-time personnel identification and compliance monitoring in secure operational environments. The primary objective of the proposed framework is to automatically detect human workers in CCTV video streams, identify visible wearable identification numbers printed on industrial uniforms or shirts, and generate automated alerts whenever a person without a visible identification number is detected within a monitored area. The system is specifically intended for deployment in high-security industrial zones, manufacturing facilities, restricted operational environments, and mission-critical infrastructures where workforce identification and safety compliance are essential.

Conventional surveillance systems primarily depend on manual monitoring of CCTV feeds, making large-scale workforce supervision inefficient, time-consuming, and prone to human error. In addition, traditional OCR-based number recognition systems frequently generate false detections by identifying unrelated numerical patterns from environmental text, machinery labels, vehicle numbers, or background signboards. To overcome these limitations, the proposed framework adopts a human-centric multi-stage computer vision pipeline in which wearable identification number recognition is performed exclusively within detected human regions. This significantly improves contextual accuracy while reducing false positives caused by irrelevant background information.

The proposed architecture integrates modern deep learning models within a modular real-time inference framework. Human detection is performed using the YOLOv8n object detection model due to its lightweight architecture, low inference latency, and suitability for continuous CCTV stream analysis. Once human subjects are localized, the corresponding regions of interest are extracted and passed to a dedicated wearable number detection model trained specifically for recognizing visible personnel identification numbers. The detected identification number is then associated with the corresponding worker and overlaid onto the processed video stream using annotated bounding boxes.

A major contribution of the proposed system is its automated compliance monitoring capability. If a detected individual does not contain a visible wearable identification number, the framework automatically generates on-screen alerts indicating the presence of a potentially unauthorized or non-compliant person. The system additionally captures evidence snapshots, maintains real-time alert logs, and supports CSV-based export functionality for incident tracking and administrative review. The framework is implemented using Python, OpenCV, Ultralytics YOLO, and CustomTkinter for both backend processing and interactive visualization. The developed graphical user interface supports video upload, live CCTV/RTSP stream integration, real-time monitoring of original and AI-processed feeds, runtime analytics, alert management, and automated MP4 output generation.

Overall, DetectWear provides a practical and scalable solution for intelligent industrial workforce monitoring and personnel compliance verification using CCTV surveillance systems. By integrating human detection, wearable identification recognition, automated alert generation, evidence logging, and real-time visualization within a unified framework, the project establishes a robust foundation for next-generation AI-powered industrial surveillance and security monitoring applications.

TABLE OF CONTENTS

Topic	Pg. No.
Bonafide Certificate	2
Declaration	3
Acknowledgement	4
Abstract	5
Table of Contents	6
Chapter 1: INTRODUCTION	8
1.1 Literature Review	11
Chapter 2: PROJECT FLOW	13
2.1 Overall System Pipeline	13
2.2 Week 1: Dataset Engineering & Problem Formulation	13
2.3 Week 2: Human Detection Model Development	14
2.4 Week 3: Wearable Number Detection Model Development	14
2.5 Week 4: GUI Development & Compliance Monitoring	15
2.6 Week 5: System Integration, Testing & Validation	16
Chapter 3: DATASET DESCRIPTION	17
3.1 Description	17
3.2 Characteristics and Integration	18
Chapter 4: BACKGROUND AND RELATED WORK	20
4.1 Selection of YOLOv8n for Real-Time Human Detection	20
4.2 Selection of YOLO11n for Wearable Identification Number Recognition	21
4.3 Selection of Deep Learning-Based Human-Centric Surveillance Pipeline	22
Chapter 5: IMPLEMENTATION	23

5.1 System Initialization	24
5.2 Dataset Preparation and Augmentation	24
5.3 Human Detection Pipeline	25
5.4 Wearable Identification Number Recognition	25
5.5 Compliance Verification and Alert Generation	26
5.6 Evidence Logging and Alert Management	27
5.7 GUI Visualization and Output Generation	28
5.8 Optional Runtime Control Mechanism	28
Chapter 6: RESULTS & DISCUSSION	32
6.1 Training and Validation Performance Analysis	33
6.2 Precision-Recall Analysis	34
6.3 Precision and Recall Evaluation	34
6.4 F1-Score Performance Analysis	36
6.5 Confusion Matrix Analysis	36
6.6 App UI for the Project	39
Chapter 7: CONCLUSION	43
Chapter 8: FUTURE SCOPE	44
References	45
Appendix	46

Chapter 1

INTRODUCTION

The rapid advancement of intelligent surveillance technologies and industrial automation systems has significantly increased the demand for real-time monitoring solutions capable of ensuring workforce safety, operational accountability, and security compliance within critical industrial environments. Modern infrastructures such as manufacturing facilities, assembly plants, restricted laboratories, mission-critical operational zones, and high-security industrial complexes continuously generate massive volumes of surveillance data through CCTV systems. Monitoring such environments manually is highly inefficient, labor-intensive, and prone to human error, especially in large-scale facilities where multiple workers operate simultaneously under strict safety and authorization protocols. Consequently, there exists a growing requirement for automated computer vision systems capable of intelligently analyzing surveillance footage, identifying personnel, and detecting operational non-compliance in real time.

In many industrial and secure operational environments, workers are required to wear uniforms containing visible identification numbers printed on shirts, jackets, or safety wear. These wearable identifiers are essential for workforce monitoring, personnel verification, restricted-area access control, operational accountability, and safety regulation enforcement. Supervisors and security personnel rely on these identifiers to ensure that only authorized workers are permitted within designated operational zones. However, conventional surveillance systems lack the capability to automatically detect and verify such wearable identification markers from live CCTV streams. As a result, workforce supervision still depends heavily on manual observation, which introduces delays in incident detection, increases monitoring complexity, and reduces overall operational efficiency.

Traditional approaches for numerical recognition commonly rely on Optical Character Recognition (OCR) techniques. While OCR systems perform effectively in structured environments with clearly visible textual information, they exhibit poor reliability in unconstrained surveillance scenarios where industrial environments contain complex backgrounds, varying illumination, motion blur, occlusions, perspective distortions, and multiple unrelated numerical patterns. CCTV footage often contains machinery labels, warning signs, vehicle registration numbers, equipment markings, and environmental text that may be incorrectly identified as personnel identifiers. Conventional OCR-based systems process the entire frame without contextual understanding, resulting in frequent false-positive detections and unnecessary computation on irrelevant regions. Such limitations significantly reduce reliability in practical industrial surveillance deployments where contextual precision and operational accuracy are critical.

To address these challenges, this project presents DetectWear, an intelligent deep learning-based industrial surveillance framework developed for real-time human-centric wearable identification number detection and compliance monitoring using CCTV streams. The primary objective of the proposed system is to automatically detect human workers within surveillance footage, identify visible wearable identification numbers exclusively from detected human regions, and generate automated alerts whenever a person without a visible identification number is detected inside the monitored environment. By restricting numerical analysis only to human subjects, the framework significantly improves contextual understanding while reducing false detections caused by irrelevant environmental text and surrounding numerical information.

The proposed framework follows a hierarchical multi-stage computer vision architecture integrating human detection, wearable number recognition, compliance verification, alert generation, evidence logging, and real-time visualization within a unified surveillance pipeline. Initially, human detection is performed using the YOLOv8n object detection architecture due to its lightweight design, low computational complexity, and fast real-time inference capability. YOLOv8n efficiently localizes human subjects within CCTV frames while maintaining high throughput performance suitable for continuous industrial surveillance applications. Once human regions are detected, the corresponding regions of interest are extracted and forwarded to a dedicated wearable identification number recognition module based on YOLO11n. The YOLO11n model is specifically trained for fine-grained wearable number detection and recognition from industrial uniforms and worker apparel. By combining YOLOv8n for efficient human localization and YOLO11n for accurate wearable identifier recognition, the proposed framework achieves a balanced trade-off between detection speed, contextual accuracy, and real-time operational performance.

A major contribution of the proposed system is its automated compliance monitoring capability. In secure industrial environments, workers are expected to visibly display their identification markers at all times while operating within restricted areas. Failure to wear such identifiers may indicate operational non-compliance, unauthorized personnel presence, or violation of workplace safety regulations. To address this requirement, the proposed framework continuously verifies the presence of visible wearable identification numbers for every detected worker. If a detected individual does not contain a visible identification number, the system automatically generates real-time on-screen alerts indicating the presence of a potentially unauthorized or non-compliant person within the monitored environment. This functionality transforms the framework from a conventional object detection application into an intelligent industrial workforce compliance monitoring platform capable of assisting security personnel and supervisors in maintaining operational discipline and safety standards.

In addition to automated alert generation, the framework incorporates evidence management and incident logging mechanisms for administrative review and compliance auditing. Whenever a non-compliant worker is detected, the system captures evidence snapshots of the corresponding individual, stores them locally, and maintains structured alert logs containing timestamp information, frame details, and incident metadata. The framework additionally supports CSV-based export functionality for incident reporting, record maintenance, and workforce compliance analysis. Such capabilities are particularly valuable in industrial environments where safety monitoring, personnel accountability, and security documentation form critical components of operational management systems.

The proposed system is implemented using Python as the primary development language, with OpenCV utilized for video processing and frame manipulation operations. Ultralytics YOLO is employed for both human detection and wearable number recognition tasks, while CustomTkinter is used to develop the graphical user interface for interactive monitoring and operational control. The developed interface provides an integrated surveillance dashboard enabling users to upload prerecorded videos, connect live CCTV or RTSP streams, dynamically start or stop AI inference, monitor real-time analytics, manage alerts, and save annotated output videos in MP4 format. The interface additionally supports dual-feed visualization, displaying both the original surveillance stream and the AI-processed output simultaneously for operational transparency and analysis.

The monitoring dashboard further displays real-time operational statistics such as frames per second (FPS), detected personnel count, identified wearable numbers, compliance alert count, runtime

statistics, and output processing status. A dedicated alert management panel displays live alert notifications along with captured evidence thumbnails, enabling operators to immediately review workforce compliance violations. These features collectively improve the usability, practicality, and industrial applicability of the proposed framework within real-world surveillance environments.

One of the major challenges associated with industrial CCTV systems is the highly unconstrained nature of surveillance footage. Industrial environments frequently exhibit difficult imaging conditions including low-resolution video streams, varying illumination, motion blur, worker occlusions, crowded operational scenes, long-distance camera placement, and varying worker orientations. Such conditions significantly affect the visibility and recognition of wearable identification numbers. To improve robustness under these challenging conditions, the wearable number recognition model is trained using augmented datasets containing variations in brightness, contrast, blur, noise, scaling, and perspective distortions commonly encountered in real-world industrial surveillance scenarios. These augmentation strategies improve the model's generalization capability and enable reliable performance under practical deployment conditions.

Another important consideration in real-time surveillance applications is computational efficiency. Industrial monitoring systems require continuous frame-by-frame processing with minimal latency to ensure timely detection and rapid alert generation. Therefore, the proposed framework prioritizes lightweight and optimized architectures capable of maintaining real-time inference performance without requiring excessive computational resources. The modular design of the framework additionally enables independent optimization of the human detection, wearable number recognition, compliance verification, visualization, and alert management modules without affecting the overall surveillance pipeline.

Unlike conventional surveillance systems that merely record footage for post-event analysis, the proposed framework actively performs intelligent scene understanding and operational compliance verification in real time. By integrating human detection, wearable identification recognition, automated alert generation, evidence logging, and real-time visualization within a unified architecture, the system establishes a practical foundation for next-generation industrial surveillance applications. The proposed framework not only improves workforce accountability and operational monitoring but also assists organizations in enforcing safety regulations, reducing unauthorized access risks, and enhancing situational awareness within secure industrial environments.

Overall, DetectWear demonstrates a scalable and practical approach for AI-powered industrial personnel identification and compliance monitoring using CCTV surveillance systems. Through the integration of deep learning-based human detection, context-aware wearable number recognition, automated compliance verification, evidence management, and real-time operational monitoring capabilities, the project establishes a robust intelligent surveillance solution suitable for modern industrial and mission-critical operational infrastructures.

1.1 Literature Review

Across the surveyed literature, several studies have explored industrial surveillance, worker monitoring, and safety compliance using deep learning-based object detection frameworks such as YOLO, Faster R-CNN, and SSD. Research works including YOLO-Based Safety Helmet Detection Systems, Real-Time PPE Compliance Monitoring using YOLOv8, Industrial Worker Safety Monitoring using Computer Vision, and Deep Learning-Based CCTV Surveillance Systems primarily focus on helmet detection, PPE verification, attendance monitoring, and restricted-area surveillance. However, most existing systems fail to address personnel identification using wearable numerical identifiers printed on worker uniforms. In addition, OCR-based approaches frequently generate false detections from machinery labels, environmental text, and unrelated background numerical patterns due to lack of contextual understanding. Furthermore, currently available frameworks rarely incorporate automated compliance verification, intelligent alert generation, evidence logging, and operational analytics required for practical industrial deployment. These limitations highlight the need for a robust human-centric surveillance framework capable of accurately detecting workers, recognizing wearable identification numbers, and generating automated alerts for non-compliant personnel in real time.

Table 1.1: Detailed Literature Review

Paper (Name + Year)	Problem Statement	Solution + Methodology + Metrics	Research Gaps
Football Jersey Number Detection using Computer Vision (2022) [1]	Difficulty in identifying jersey numbers from low-resolution sports footage. .	Uses CNN + OCR pipeline for player jersey number recognition from broadcast videos.	Poor robustness under occlusion, motion blur, and non-sports industrial environments.
Jersey Number Detection using YOLO and OCR (2023) [2]	Detecting player identification numbers from video streams.	Uses YOLO for player localization and OCR for number extraction.	OCR generates false detections and struggles with complex backgrounds.
Soccer Jersey Number Recognition Using Convolutional Neural Networks (2015) [3]	Difficulty in recognizing jersey numbers from low-resolution sports footage.	Uses CNN-based classification for jersey number recognition from player regions in broadcast videos.	Limited robustness under occlusion, blur, and real-world surveillance environments.
Jersey Number Recognition With Semi-Supervised Spatial Transformer Network (2018) [4]	Accurate jersey number recognition under varying player orientations and scales.	Uses Spatial Transformer Networks with semi-supervised learning for number localization and recognition.	Computational complexity and limited applicability outside sports datasets.
Pose-Guided R-CNN for Jersey Number Recognition in Sports (2019) [5]	Detecting jersey numbers under pose variation and crowded scenes.	Uses pose-guided R-CNN with player keypoint estimation for improved jersey localization.	Requires complex pose estimation and struggles under severe occlusion.

Paper (Name + Year)	Problem Statement	Solution + Methodology + Metrics	Research Gaps
Knock, Knock. Who's There? Identifying Football Player Jersey Numbers with Synthetic Data (2022) [6]	Limited availability of annotated jersey datasets for deep learning training.	Uses synthetic dataset generation with CNN-based jersey recognition pipeline.	Focused mainly on sports analytics and lacks compliance monitoring features.
Jersey Number Recognition using Keyframe Identification from Low-Resolution Broadcast Videos (2023) [7]	Poor jersey visibility in low-resolution sports broadcasts.	Uses keyframe extraction and deep learning-based recognition for improved number detection.	Limited real-time capability and no contextual worker verification.
A General Framework for Jersey Number Recognition in Sports Video (2024) [8]	Need for generalized jersey number recognition across multiple sports datasets.	Uses multi-stage deep learning framework for robust number localization and classification.	Focused only on sports scenarios and lacks industrial surveillance integration.
Real-Time Face Detection & Recognition in CCTV (2022) [9]	Need for automated surveillance without human monitoring.	Uses PCA + CNN + KNN + Random Forest. Dataset of 40K images. Achieves >90% accuracy.	Sensitive to lighting, occlusion; not optimized for real-time edge systems; lacks interpretability.
Real-Time Robust Human Detection and Tracking System (2005) [10]	Difficulty in reliable human detection in dynamic environments (lighting changes, shadows, background motion).	Uses background subtraction + Gaussian modeling + codebook classification. Tracking via color histogram. Includes false alarm detection module. Works in real-time environments.	Poor performance in highly dynamic scenes; limited adaptability to complex backgrounds; no deep learning; weak occlusion handling.

Chapter 2

PROJECT FLOW

2.1 Overall System Pipeline

1. CCTV/RTSP Video Stream Acquisition
2. Frame Extraction & Video Preprocessing
3. Human Detection using YOLOv8n
4. Human Region Extraction using Bounding Box Cropping
5. Wearable Number Detection & Identification using YOLO11n
6. Compliance Verification for Number Presence
7. Real-Time No-Number Alert Generation
8. Alert Logging & Evidence Image Capture
9. Annotated Video Generation with Bounding Boxes & ID Labels
10. GUI Visualization with Original and AI-Processed Feeds
11. Runtime Analytics (FPS, Human Count, Alert Count, Detection Status)
12. Output MP4 Video Saving & CSV Alert Export

2.2 Week 1: Dataset Engineering & Problem Formulation

Table 2.1: Week 1 Workflow

Component	Tasks	Technical Details	Deliverables
Problem Definition	Define industrial surveillance objective	Human-centric wearable number detection for secure industrial environments	Formal problem statement
Human Detection Planning	Model selection for personnel detection	YOLOv8n selected for lightweight real-time human detection	Human detection framework
Wearable Number Dataset Collection	Dataset preparation for wearable number detection	Roboflow-based wearable/jersey number dataset collection	Raw wearable dataset
Dataset Organization	Folder structuring and annotation setup	YOLO-compatible train/validation dataset organization	Structured dataset
Annotation Verification	Bounding-box validation	Validation of wearable number annotations and label consistency	Verified annotations
Augmentation	Robustness enhancement	Flipping, brightness variation, minor affine transforms	Augmented dataset
Training Setup	GPU environment configuration	CUDA-enabled PyTorch and Ultralytics setup	Training environment

2.3 Week 2: Human Detection Model Development

Table 2.2: Week 2 Workflow

Component	Tasks	Technical Details	Deliverables
Human Dataset Preparation	Person-class dataset setup	COCO person-class subset preparation	Human dataset
YOLOv8n Configuration	Human detection model setup	Lightweight real-time detection architecture	Configured YOLOv8n
Model Training	Human detection training	GPU-accelerated YOLOv8n training	Trained human detector
Bounding Box Optimization	Detection refinement	Confidence threshold tuning	Optimized detections
Frame Processing	Video inference setup	OpenCV-based frame-by-frame processing	Video processing module
ROI Extraction	Human region cropping	Bounding-box-based human extraction	ROI extraction pipeline
Validation Testing	Performance verification	Human detection testing on surveillance videos	Validated human detector

2.4 Week 3: Wearable Number Detection Model Development

Table 2.3: Week 3 Workflow

Component	Tasks	Technical Details	Deliverables
YOLO11n Setup	Wearable detection configuration	YOLO11n selected for wearable number detection	Configured YOLO11n
Dataset Training	Model training	Roboflow dataset trained for 80 epochs	Trained wearable detector
Number Localization	Wearable number detection	Bounding-box-based number recognition	Number detection module
Human-Centric Filtering	Context-aware processing	Number recognition only within human regions	Reduced false positives

Robustness Enhancement	Model optimization	Blur and illumination variation handling	Robust detection model
Detection Validation	Accuracy testing	Testing on multiple video frames	Verified number detector
Pipeline Integration	Model connection	YOLOv8n + YOLO11n integration	Integrated AI pipeline

2.5 Week 4: GUI Development & Compliance Monitoring

Table 2.4: Week 4 Workflow

Component	Tasks	Technical Details	Deliverables
GUI Development	Interface design	CustomTkinter-based surveillance dashboard	Interactive GUI
Dual Feed Visualization	Video display setup	Original and AI-processed feed visualization	Dual-feed system
Alert Generation	Compliance monitoring	No-number alert generation system	Alert engine
Evidence Capture	Incident recording	Worker snapshot capture during alerts	Evidence module
CCTV Integration	Live stream connectivity	RTSP and CCTV stream support	CCTV monitoring
Output Saving	MP4 output generation	OpenCV video writer integration	Saved surveillance videos
Runtime Analytics	Dashboard statistics	FPS, human count, and alert monitoring	Analytics panel

2.6 Week 5: System Integration, Testing & Validation

Table 2.5: Week 5 Workflow

Component	Tasks	Technical Details	Deliverables
Full Pipeline Integration	End-to-end framework connection	YOLOv8n + YOLO11n + GUI integration	Complete AI system
Real-Time Testing	Surveillance validation	Testing on prerecorded and CCTV streams	Tested surveillance pipeline
Alert Validation	Compliance verification	No-number alert testing	Verified alert system
Multi-Person Testing	Crowded-scene evaluation	Simultaneous worker detection testing	Multi-worker validation
Output Verification	Annotation checking	Bounding-box and label verification	Verified AI output
CSV Export Testing	Alert report validation	CSV-based incident export testing	Alert reports
Final Optimization	Performance refinement	GUI responsiveness and inference optimization	Finalized DetectWear framework

Chapter 3

DATASET DESCRIPTION

3.1 Description

The dataset design follows a two-stage construction paradigm aligned with the dual objectives of:

- (i) **robust human localization in industrial surveillance environments, and**
- (ii) accurate **wearable identification number detection from worker uniforms** using deep learning-based object detection architectures.

The overall dataset pipeline is divided into two independent yet interconnected stages corresponding to the human detection module and wearable number recognition module within the proposed surveillance framework.

Stage 1: Human Detection-Oriented Dataset (COCO Subset Filtering)

For the human detection task, a selectively filtered subset of the COCO dataset is utilized as the primary source of supervisory signals for training and validating the YOLOv8n detection architecture. The dataset is filtered specifically to retain only the person class annotations, enabling the model to focus exclusively on human localization within surveillance frames. Bounding-box annotations are utilized to generate spatial priors for human detection under varying environmental conditions including crowd density, pose variation, occlusion, illumination changes, and complex industrial backgrounds.

The YOLOv8n architecture is employed as the primary human detection backbone due to its lightweight structure, low computational complexity, and suitability for real-time inference within CCTV surveillance systems. Since the proposed framework focuses specifically on human-centric wearable identification analysis, accurate human localization forms a critical preprocessing stage for downstream wearable number recognition.

Preprocessing operations performed on the human detection dataset include:

- Spatial normalization
- Resolution scaling
- Aspect-ratio preservation
- Bounding-box consistency verification
- RGB image normalization

These preprocessing steps ensure compatibility with the YOLOv8n training pipeline while preserving spatial integrity and detection robustness under unconstrained industrial surveillance conditions.

Stage 2: Wearable Identification Number Dataset

To facilitate wearable identification number recognition, the proposed system utilizes a Roboflow-based wearable number detection dataset containing annotated numerical identifiers printed on jerseys, uniforms, and worker apparel. The dataset is specifically structured for object detection-based numerical localization rather than OCR-based text extraction. Unlike traditional OCR systems that process entire frames indiscriminately, the proposed framework performs wearable number detection only within detected human

regions, significantly improving contextual precision and reducing false-positive detections from environmental numerical patterns.

The wearable number dataset contains annotated bounding boxes corresponding to visible numerical identifiers under varying conditions such as:

- Illumination variation
- Motion blur
- Scale variation
- Camera distance variation
- Perspective distortion
- Partial occlusion
- Low-resolution CCTV imagery
-

The YOLO11n architecture is utilized for wearable number recognition due to its efficient detection capability and suitability for fine-grained object localization tasks. The model is trained for 80 epochs using GPU-accelerated training with CUDA-enabled PyTorch and Ultralytics YOLO frameworks. Since the system relies entirely on deep learning-based object detection, no OCR-based textual extraction techniques are incorporated within the pipeline.

To improve model generalization and robustness under real-world surveillance conditions, multiple augmentation strategies are applied during training, including:

- Brightness and contrast variation
- Motion blur simulation
- Random scaling
- Horizontal transformations
- Noise injection
- Low-light enhancement
- Minor perspective distortions

These augmentation techniques improve the ability of the wearable number detection model to operate reliably under practical industrial CCTV environments where worker visibility and number clarity may vary significantly.

3.2 Characteristics and Integration

The resulting dataset exhibits a hybrid structural composition combining large-scale human detection annotations with fine-grained wearable identification number localization. This decoupled dataset design enables the framework to independently optimize:

- Human localization using YOLOv8n
- Wearable number recognition using YOLO11n

Integration is achieved through a sequential inference pipeline in which detected human regions are extracted from surveillance frames and subsequently forwarded to the wearable number recognition module. This human-centric inference strategy significantly reduces false detections from irrelevant environmental numerical patterns such as machinery labels, warning boards, vehicle numbers, and industrial equipment markings.

The dataset design inherently supports real-time surveillance deployment by enabling efficient frame-level processing, lightweight inference, and modular scalability. Since the wearable number recognition stage operates only on cropped human regions, the overall computational overhead is significantly reduced compared to conventional frame-wide OCR-based systems.

Furthermore, the dataset pipeline is incrementally extensible, allowing new wearable identification classes and additional worker samples to be incorporated without redesigning the overall surveillance architecture. This flexibility enables the framework to adapt to evolving industrial deployment scenarios involving varying worker uniforms, operational conditions, and surveillance environments.

Overall, the dataset provides a robust foundation for a multi-stage intelligent surveillance system capable of performing human detection, wearable identification number recognition, automated compliance verification, alert generation, and real-time CCTV monitoring under practical industrial conditions.

Chapter 4

BACKGROUND & RELATED WORK

4.1 Selection of YOLOv8n for Real-Time Human Detection

- **Optimal speed–accuracy trade-off:** YOLOv8n provides a validation mAP of 37.3 while maintaining a significantly lower inference time (80.4 ms on CPU) compared to larger variants like YOLOv8m (234.7 ms) and YOLOv8x (479.1 ms). This makes it suitable for real-time applications where latency is critical.
- **Real-time video processing capability:** The reduced inference time of YOLOv8n enables higher frame rates (~12 FPS on CPU), whereas larger models drop to impractical speeds (~2–4 FPS), making them unsuitable for continuous CCTV/video stream processing.
- **Lightweight architecture:** With only 3.2 million parameters and 8.7 GFLOPs, YOLOv8n is computationally efficient compared to YOLOv8x (68.2M parameters, 257.8 GFLOPs), allowing deployment on resource-constrained systems such as edge devices or standard CPUs.
- **System-level efficiency:** The proposed pipeline already incorporates computationally intensive modules like MTCNN and FaceNet for face detection and recognition. Using a lightweight detector (YOLOv8n) ensures that the overall system remains efficient without bottlenecking the processing pipeline.
- **Task-specific adequacy:** Since the objective is limited to human detection, the marginal accuracy improvement offered by larger models (e.g., YOLOv8x achieving 53.9 mAP) does not justify the substantial increase in computational cost and latency.
- **Scalability and deployment feasibility:** YOLOv8n’s low memory footprint and faster inference make it more suitable for scalable deployment in real-world environments, including embedded systems and live surveillance setups.

Table 4.1: Comparison of YOLOv8 Model Variants Based on Performance and Complexity

Model	Size (pixels)	mAP (50–95)	Speed CPU ONNX (ms)	Speed A100 TensorRT (ms)	Params (M)	FLOPs (B)
YOLOv8n	640	37.3	80.4	0.99	3.2	8.7
YOLOv8s	640	44.9	128.4	1.20	11.2	28.6
YOLOv8m	640	50.2	234.7	1.83	25.9	78.9
YOLOv8l	640	52.9	375.2	2.39	43.7	165.2
YOLOv8x	640	53.9	479.1	3.53	68.2	257.8

4.2 Selection of YOLO11n for Wearable Identification Number Recognition

- **Fine-grained object detection capability:** YOLO11n provides improved detection performance for small and fine-grained objects such as wearable identification numbers present on worker uniforms.
- **Real-time number recognition:** The lightweight architecture enables continuous wearable number recognition from CCTV streams without significant computational delay.
- **Human-centric contextual detection:** The proposed framework performs wearable number detection only within detected human regions, allowing YOLO11n to focus specifically on worker-associated identifiers rather than irrelevant background information.
- **Improved robustness under industrial conditions:** YOLO11n demonstrates strong performance under challenging surveillance conditions including motion blur, illumination variation, low-resolution imagery, and partial occlusion.
- **Pure deep learning-based recognition:** Unlike OCR-based systems that rely on textual extraction, YOLO11n directly performs object detection-based wearable number localization, reducing false detections from environmental text and machinery labels.
- **Efficient GPU-based training:** The wearable number detection model was trained using GPU-accelerated CUDA-based training on the Roboflow dataset for 80 epochs, improving convergence stability and recognition capability.
- **Scalable deployment suitability:** YOLO11n provides a balanced trade-off between recognition accuracy and computational efficiency, making it suitable for real-time industrial surveillance applications.

Table 4.2: Comparison of NumberDetection Methods

Method	Recognition	Real-Time Capability	Robustness	Computational Complexity	False Positive Handling
OCR-Based Systems	Text Extraction	Moderate	Low	Low	Poor
CNN Classification	Image Classification	Moderate	Medium	Medium	Moderate
Faster R-CNN	Two-stage Object Detection	Low	High	High	Good
SSD	Single-stage Object Detection	High	Moderate	Medium	Moderate
YOLO11n	Single-stage Object Detection	Very High	Very High	Low	Excellent

4.3 Selection of Deep Learning-Based Human-Centric Surveillance Pipeline

- **Context-aware surveillance design:** The proposed framework performs wearable number recognition only after human detection, reducing false detections from background numerical patterns. .
- **Reduction of OCR limitations:** Traditional OCR systems frequently misidentify machinery labels, warning signs, and environmental text as valid identifiers. The proposed deep learning-based pipeline eliminates this issue through contextual filtering.
- **Improved operational reliability:** Human-centric region filtering significantly improves wearable identification precision under practical industrial conditions.
- **Automated compliance monitoring:** The framework incorporates intelligent alert generation for workers without visible wearable identification numbers.
- **Integrated evidence logging:** Real-time alert logs and evidence image capture improve incident tracking and workforce monitoring capabilities.
- **Interactive surveillance visualization:** The developed GUI enables real-time visualization of both original and AI-processed CCTV feeds with operational analytics.
- **Scalable modular architecture:** The decoupled design enables independent optimization of human detection, wearable number recognition, compliance verification, and visualization modules without affecting the complete surveillance pipeline.
- **Robustness in unconstrained environments:**
 - Improved performance under varying lighting conditions
 - Essential for **CCTV/video-based applications**

Table 4.3: Comparative Analysis of Surveillance Approaches

System Type	Human Detection	Number Recognition	Alert Generation	Context Awareness	Industrial Suitability
Traditional CCTV Monitoring	Manual	Manual	No	No	Limited
OCR-Based Surveillance	Partial	OCR	Limited	Poor	Moderate
Generic Object Detection Systems	Yes	No	No	Moderate	Moderate
PPE Monitoring Systems	Yes	No	Yes	Moderate	High
Proposed DetectWear Framework	YOLOv8n	YOLO11n	Yes	Good	High

Chapter 5

IMPLEMENTATION

The proposed DetectWear framework is implemented as a multi-stage intelligent industrial surveillance pipeline integrating real-time human detection, wearable identification number recognition, automated compliance monitoring, alert generation, evidence logging, and interactive visualization within a unified architecture. The system follows a modular design strategy, enabling independent optimization of each processing stage while maintaining end-to-end compatibility for real-time CCTV surveillance deployment. The implementation primarily utilizes Python, OpenCV, Ultralytics YOLO, PyTorch, and CustomTkinter for both backend inference and graphical visualization.

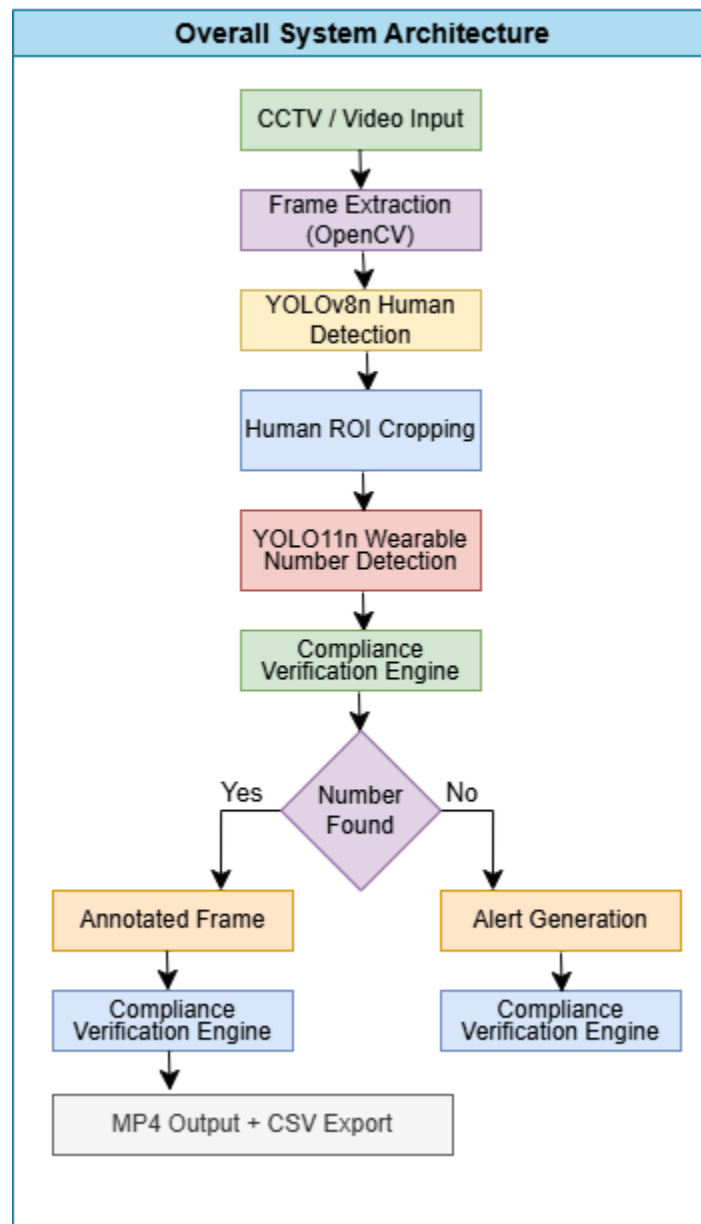


Figure 5.1: Overall System Architecture

5.1 System Initialization

The framework begins with runtime initialization, where the computation device is selected dynamically based on hardware availability. CUDA-enabled GPU acceleration is utilized whenever supported hardware is detected to enable faster inference during surveillance processing.

The system initializes the YOLOv8n human detection model, YOLO11n wearable number recognition model, OpenCV video processing pipeline, alert management system, and GUI visualization modules. Runtime parameters such as confidence thresholds, alert cooldown timers, frame-skipping settings, and output saving configurations are also initialized before execution to ensure stable real-time CCTV processing.

5.2 Dataset Preparation and Augmentation

The proposed framework utilizes two separate datasets for human detection and wearable identification number recognition. For the human detection stage, a filtered subset of the COCO dataset containing only the person class is used to train the YOLOv8n model for efficient human localization in surveillance environments.

For wearable number recognition, a Roboflow-based dataset containing annotated wearable numerical identifiers printed on worker uniforms and safety wear is utilized. The dataset is organized using a YOLO-compatible train-validation structure with corresponding annotation files.

To improve robustness under practical CCTV conditions, augmentation techniques such as motion blur, brightness variation, scaling, low-light enhancement, noise injection, and minor perspective distortions are applied during training. These augmentations simulate real-world industrial surveillance scenarios involving illumination variation, worker motion, and low-resolution imagery.

The YOLO11n wearable number detection model is trained for 80 epochs using CUDA-enabled GPU acceleration with PyTorch and Ultralytics YOLO frameworks.

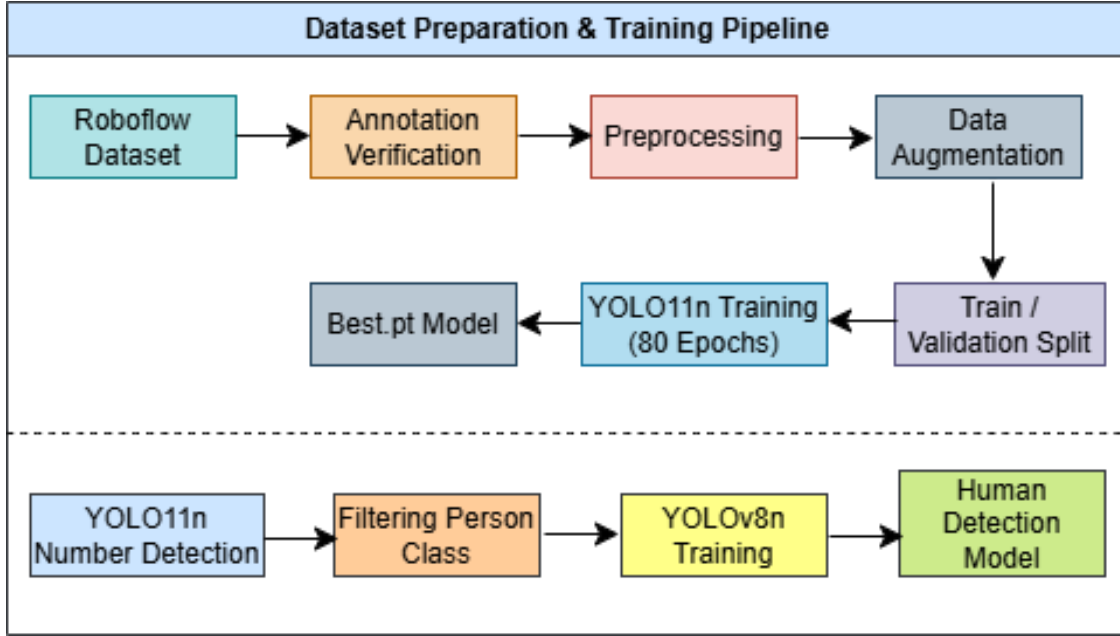


Figure 5.2: Face Dataset Preparation and Augmentation Pipeline

5.3 Human Detection Pipeline

The system processes incoming video streams frame-by-frame using OpenCV-based capture mechanisms. Each frame is forwarded to the YOLOv8n object detection model for real-time human localization.

The detector outputs:

- Bounding box coordinates
- Confidence scores
- Human class predictions

Low-confidence detections are discarded using predefined confidence thresholds to improve reliability and reduce false-positive detections. Only valid human regions are selected for downstream wearable number analysis.

This stage acts as the primary contextual filtering mechanism within the surveillance pipeline by ensuring that wearable identification analysis is restricted exclusively to detected human subjects rather than the entire frame.

5.4 Wearable Identification Number Recognition

For every detected human subject, the corresponding bounding-box region is extracted and forwarded to the YOLO11n wearable identification recognition module. Unlike OCR-based systems that directly perform text extraction, the proposed framework uses pure deep learning-based object detection for recognizing wearable numerical identifiers.

The YOLO11n model detects:

- Visible numerical identifiers
- Wearable number bounding boxes
- Number confidence scores

The detected wearable number is then associated with the corresponding worker and displayed directly on the processed surveillance feed using annotated labels and bounding boxes.

This human-centric number recognition strategy significantly reduces false-positive detections from:

- Machinery labels
- Warning boards
- Environmental text
- Vehicle numbers
- Industrial equipment markings

thereby improving contextual reliability within industrial environments.

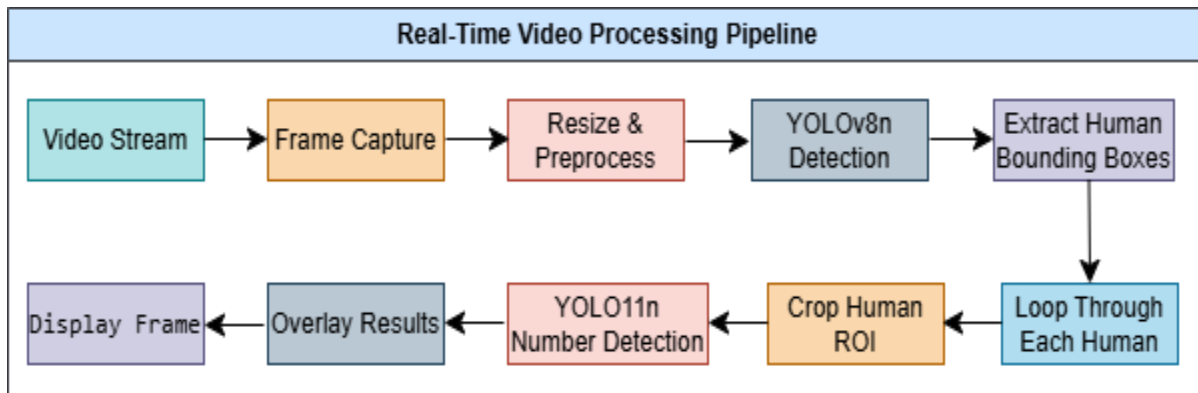


Figure 5.3: Real-Time Video Processing Pipeline

5.5 Compliance Verification and Alert Generation

The framework incorporates an automated compliance monitoring mechanism to identify workers without visible wearable identification numbers. After wearable number recognition, the system verifies whether every detected human contains a valid numerical identifier.

If no wearable number is detected for a worker:

- The worker is marked as non-compliant
- A real-time on-screen alert is generated
- Alert bounding boxes are displayed
- Alert logs are updated

To prevent repeated alerts for the same worker within short durations, cooldown timers and frame-based persistence mechanisms are implemented. The system additionally maintains no-number detection streak counters to improve alert stability and reduce false triggering caused by temporary detection failures.

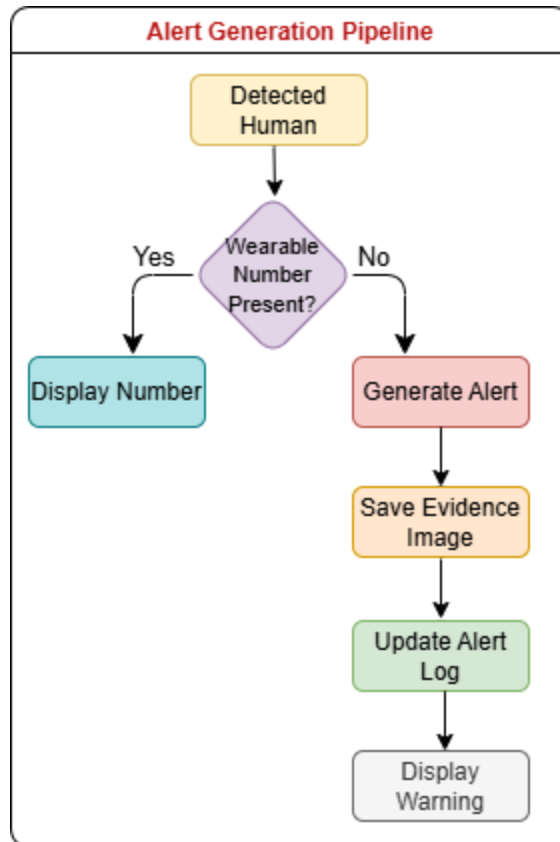


Figure 5.4: Alert Generation Pipeline

5.6 Evidence Logging and Alert Management

Whenever a compliance violation occurs, the system captures evidence snapshots of the corresponding worker and stores them locally within the alert management directory.

The framework records:

- Alert timestamp
- Frame information
- Detection status
- Worker evidence image
- Alert type

Structured CSV-based logging mechanisms are implemented to enable incident documentation and compliance reporting for administrative review.

The alert management system additionally supports:

- Live alert visualization
- Evidence thumbnail display
- Historical alert tracking
- Runtime alert analytics

through the developed graphical dashboard.

5.7 GUI Visualization and Output Generation

The proposed framework includes a CustomTkinter-based graphical interface designed for real-time industrial surveillance monitoring. The GUI provides:

- Video upload support
- CCTV/RTSP stream integration
- Dual-feed visualization
- Runtime analytics
- Alert monitoring
- Output video generation

The interface simultaneously displays:

- Original surveillance feed
- AI-processed annotated feed

Operational analytics displayed within the dashboard include:

- Frames per second (FPS)
- Human detection count
- Wearable number count
- Alert count
- Runtime statistics

The system additionally supports MP4-based output video generation where annotated surveillance footage containing human detections, wearable identifiers, and compliance alerts is saved for future analysis and playback.

5.8 Optional Runtime Control Mechanism

To maintain real-time surveillance performance, multiple runtime optimization strategies are incorporated within the framework. These include:

- Frame skipping mechanisms
- Region-of-interest processing
- Lightweight model architectures
- Detection persistence reuse
- GPU-accelerated inference

Instead of performing full inference on every frame, the system selectively performs detection at configurable intervals while reusing previous detections for intermediate frames. This significantly improves FPS performance without major degradation in detection quality.

The lightweight nature of YOLOv8n and YOLO11n further enables efficient execution on GPU-enabled systems while maintaining low latency suitable for continuous industrial CCTV monitoring applications.

```

jersey_detection_ui.py 1
jersey_detection_ui.py > JerseyDetectionUI
10 from ultralytics import YOLO
11 import torch
12
13 VIDEO_PATH = r"uploads/testvidslo.mp4"
14 HUMAN_MODEL_PATH = r"best1/best.pt"
15 JERSEY_MODEL_PATH = r"bestf1/best.pt"
16 timestamp = datetime.datetime.now().strftime("%Y%m%d_%H%M%S")
17 OUTPUT_DIR = "outputs"
18 ALERT_DIR = "alerts"
19 OUTPUT_PATH = rf"outputs/output_{timestamp}.mp4"
20 os.makedirs(OUTPUT_DIR, exist_ok=True)
21 os.makedirs(ALERT_DIR, exist_ok=True)
22 YELLOW = "#FFC300"
23 LIGHT_YELLOW = "#FFD95A"
24 BG = "#050505"
25 CARD = "#101010"
26 INNER = "#020202"
27 ALERT = "#FF3B30"
28 WHITE = "#FFFFFF"
29
30 ctk.set_appearance_mode("dark")
31 ctk.set_default_color_theme("dark-blue")
32
33 class JerseyDetectionUI(ctk.CTk):
34     def __init__(self):
35         super().__init__()
36         self.title("AI-Based Human-Centric Wearable Identification")
37         self.geometry("1400x820")
38         self.minsize(1200, 720)
39         self.configure(fg_color=BG)
40         self.video_source = VIDEO_PATH
41         self.cap = None
42         self.running = False
43         self.thread = None
44         self.human_model = None
45         self.jersey_model = None
46         self.writer = None

```

Figure 5.5: Snippet of jersey_detection_ui.py code

```
jersey_detection_ui.py  train_yolo11_digits.py 1 X
train_yolo11_digits.py > ...
1  from ultralytics import YOLO
2  import torch
3  from multiprocessing import freeze_support
4
5  if __name__ == "__main__":
6      freeze_support()
7
8      DIGIT_MODEL_PATH = r"bestf1/best.pt"
9
10     EPOCHS = 80
11     IMG_SIZE = 640
12     BATCH_SIZE = 8
13     WORKERS = 0
14
15     print("CUDA available:", torch.cuda.is_available())
16
17     if torch.cuda.is_available():
18         print("GPU:", torch.cuda.get_device_name(0))
19         DEVICE = 0
20     else:
21         DEVICE = "cpu"
22
23     model = YOLO("yolo11n.pt")
24
25     model.train(
26         data=DATA_YAML,
27         epochs=EPOCHS,
28         imgsz=IMG_SIZE,
29         batch=BATCH_SIZE,
30         device=DEVICE,
31         workers=WORKERS,
32         cache=False,
33         amp=True,
34         project="runs/jersey_digit_detector",
35         name="yolo11_digits_final",
36         exist_ok=True
37     )
```

Figure 5.6: Snippet of train_yolo11_digits.py code

```
jersey_detection_ui.py  train_yolo11_digits.py  train_yolo11_humans.py X
train_yolo11_humans.py > ...
1  from ultralytics import YOLO
2  import torch
3  from multiprocessing import freeze_support
4
5  if __name__ == "__main__":
6      freeze_support()
7
8      DATA_YAML = r"datasets/human_object_dataset/data.yaml"
9
10     EPOCHS = 75
11     IMG_SIZE = 640
12     BATCH_SIZE = 16
13     WORKERS = 4
14
15     print("CUDA available:", torch.cuda.is_available())
16
17     if torch.cuda.is_available():
18         print("GPU:", torch.cuda.get_device_name(0))
19         DEVICE = 0
20     else:
21         DEVICE = "cpu"
22
23     model = YOLO("yolov8n.pt")
24
25     model.train(
26         data=DATA_YAML,
27         epochs=EPOCHS,
28         imgsz=IMG_SIZE,
29         batch=BATCH_SIZE,
30         device=DEVICE,
31         workers=WORKERS,
32         cache=True,
33         amp=True,
34         project="runs/human_detector",
35         name="yolov8n_human_object_75epochs",
36         exist_ok=True
37     )
```

Figure 5.7: Snippet of train_yolo11_humans.py code

Chapter 6

RESULTS & DISCUSSION

The proposed DetectWear framework was successfully implemented and evaluated for real-time industrial workforce monitoring and wearable identification number recognition using CCTV surveillance streams. The system integrates YOLOv8n-based human detection with YOLO11n-based wearable number recognition to perform automated personnel monitoring, compliance verification, alert generation, evidence logging, and interactive visualization within a unified surveillance framework.

The YOLO11n wearable number recognition model was trained using a Roboflow-based dataset for 80 epochs under CUDA-enabled GPU acceleration. Experimental results demonstrated stable convergence during training with gradual reduction in training and validation losses across epochs. Simultaneously, detection metrics including precision, recall, F1-score, and mean Average Precision (mAP) consistently improved throughout the training process, indicating effective feature learning and robust generalization capability under varying industrial surveillance conditions.

The obtained Precision-Recall curves demonstrated strong detection performance across multiple numerical classes with an overall mAP@0.5 score of approximately 0.964. Similarly, the Precision-Confidence and Recall-Confidence curves illustrated balanced detection behavior across varying confidence thresholds, confirming the model's capability to maintain stable recognition performance under practical CCTV environments. The F1-Confidence analysis additionally indicated an overall F1-score of approximately 0.96, reflecting balanced precision and recall performance with reduced false-positive and false-negative detections.

The confusion matrix and normalized confusion matrix further validated the effectiveness of the wearable number recognition pipeline. Most numerical classes achieved high classification accuracy with minimal inter-class confusion. Minor misclassifications were primarily observed among visually similar numerical patterns under low-resolution and partially occluded surveillance conditions. Nevertheless, the overall recognition performance remained highly stable and suitable for real-time industrial deployment.

The integrated YOLOv8n-based human detection module successfully localized workers under varying illumination, worker movement, and multi-person surveillance scenarios. The human-centric inference strategy significantly reduced false detections from environmental numerical patterns such as machinery labels, warning boards, industrial equipment markings, and background text by restricting wearable number recognition exclusively to detected human regions.

The developed compliance verification engine successfully generated real-time alerts whenever workers without visible wearable identification numbers were detected. The framework additionally supported automated evidence image capture, alert logging, MP4-based output video generation, CSV-based incident export, and interactive GUI visualization. Runtime analytics including FPS, worker count, wearable number count, and alert statistics were continuously displayed through the developed surveillance dashboard.

Overall, the proposed DetectWear framework successfully demonstrates the feasibility of integrating deep learning-based human detection, wearable number recognition, automated

compliance monitoring, and intelligent CCTV analytics within a unified industrial surveillance system. The framework establishes a scalable and practical solution for secure workforce monitoring and real-time personnel compliance verification in industrial environments.

6.1 Training and Validation Performance Analysis

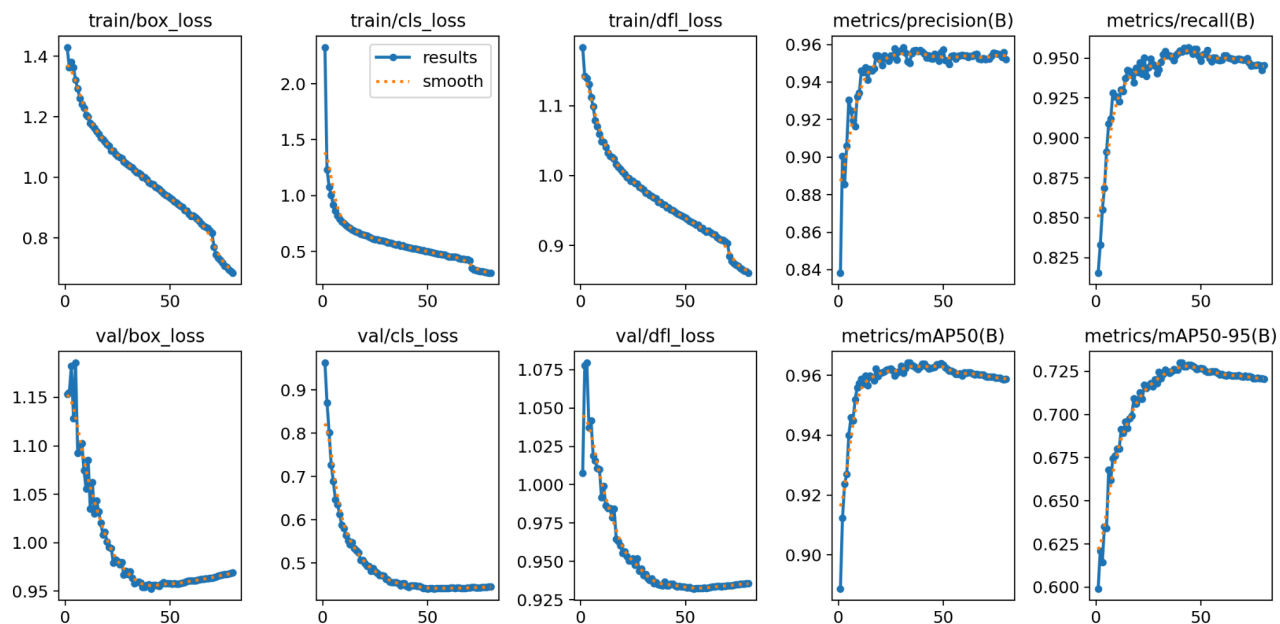


Figure 6.1: Training and Validation Performance Metrics of YOLO11n Wearable Number Detection Model

The YOLO11n wearable identification number recognition model was trained for 80 epochs using the Roboflow-based wearable number dataset under CUDA-enabled GPU acceleration. Figure 6.1 illustrates the training and validation performance metrics obtained during the training process. The training box loss, classification loss, and distribution focal loss progressively decreased across epochs, indicating stable model convergence and effective optimization during training.

Similarly, the validation losses demonstrated consistent reduction without significant overfitting, confirming the generalization capability of the proposed wearable number recognition model under varying surveillance conditions. In addition, the precision, recall, mAP@0.5, and mAP@0.5:0.95 metrics steadily improved throughout training, demonstrating enhanced detection accuracy and reliable localization performance across multiple wearable number classes.

The model achieved high precision and recall values along with stable mAP performance, indicating the effectiveness of the YOLO11n architecture for fine-grained wearable number recognition in real-time industrial CCTV environments. The gradual convergence of both training and validation metrics further confirms the robustness and stability of the proposed deep learning-based detection framework.

6.2 Precision-Recall Analysis

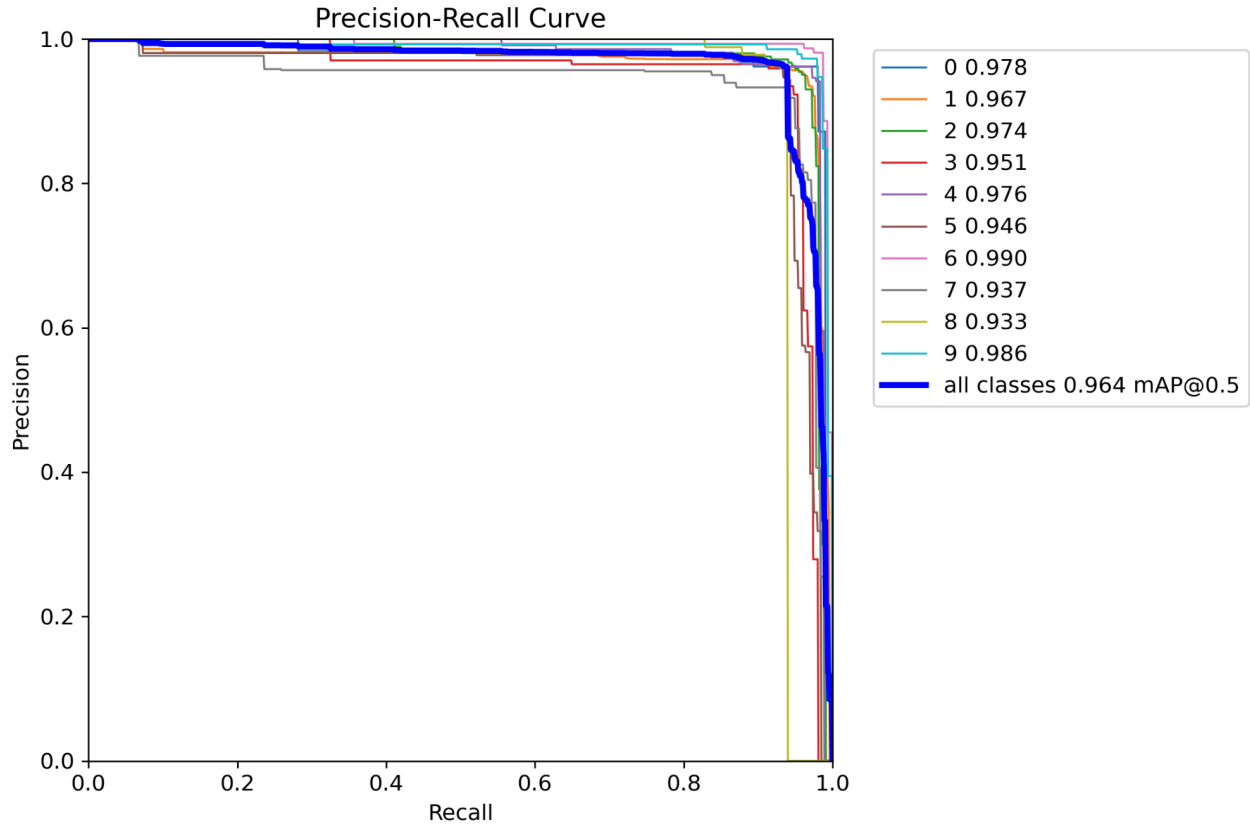


Figure 6.2: Precision-Recall Curve for Wearable Number Recognition

The Precision-Recall curve shown in Figure 6.2 demonstrates the detection capability of the proposed wearable number recognition model across multiple numerical classes. The model achieved high precision values even at higher recall levels, indicating strong classification reliability and reduced false-positive detections.

The overall mAP@0.5 value achieved by the model was approximately 0.964, demonstrating robust wearable number localization and recognition performance across multiple surveillance scenarios.

6.3 Precision and Recall Evaluation

Figures 6.3 and 6.4 illustrate the Precision-Confidence and Recall-Confidence curves respectively for the wearable number recognition model. The precision curve demonstrates stable precision performance across varying confidence thresholds, while the recall curve indicates effective retrieval capability for wearable numerical identifiers.

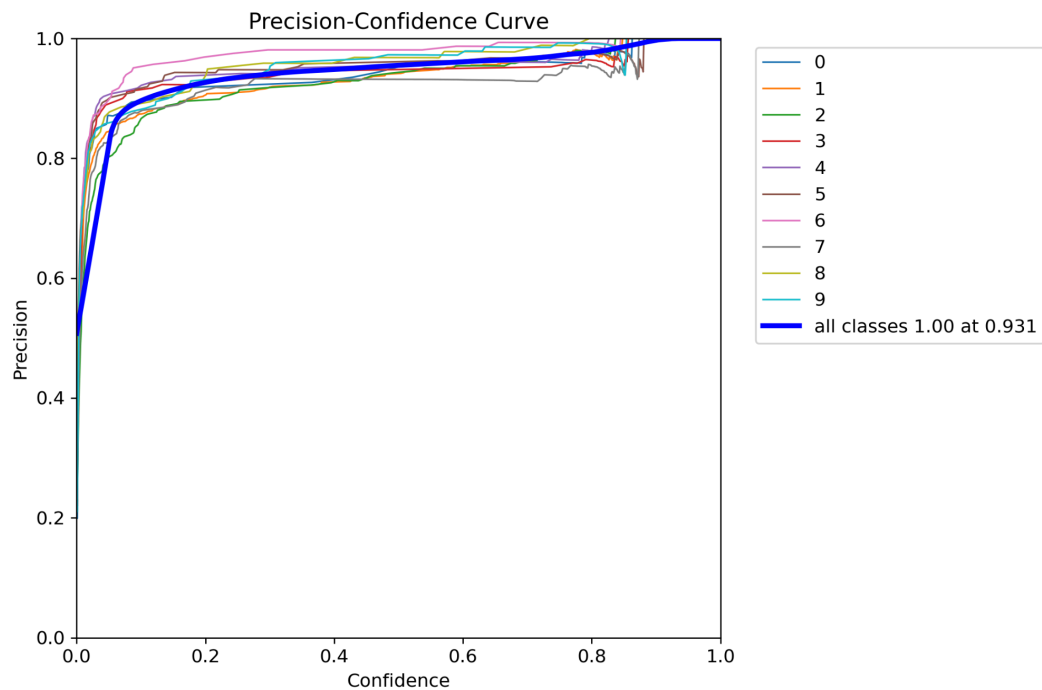


Figure 6.3: Precision-Confidence Curve

The experimental results demonstrate that the proposed YOLO11n-based framework maintains balanced precision and recall performance suitable for real-time industrial surveillance applications.

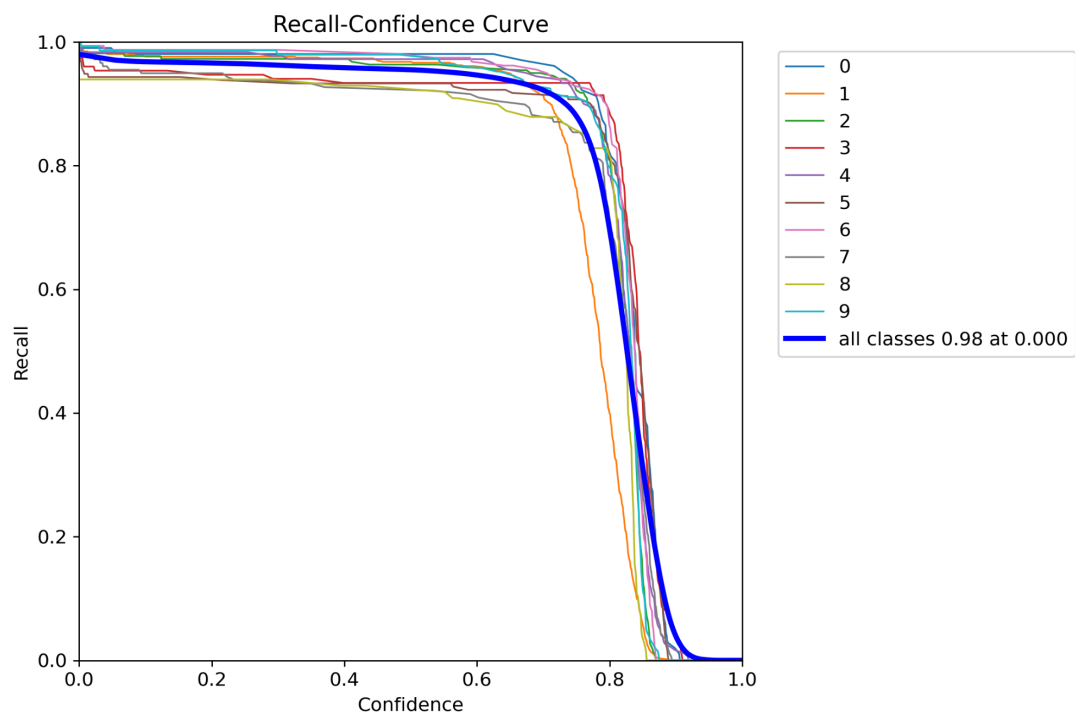


Figure 6.4: Recall-Confidence Curve

6.4 F1-Score Performance Analysis

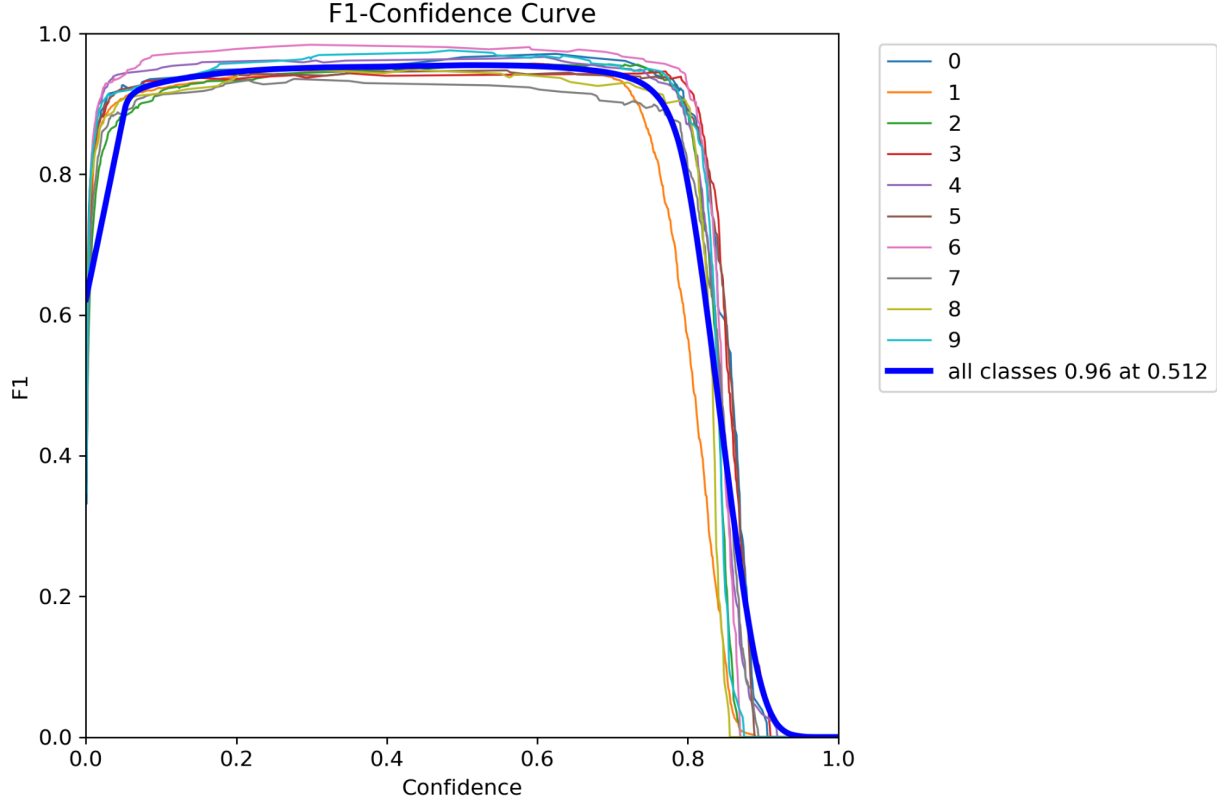


Figure 6.5 : F1-Confidence Curve for Wearable Number Recognition

The F1-Confidence curve shown in Figure 6.5 represents the harmonic balance between precision and recall across varying confidence thresholds. The model achieved an overall F1-score of approximately **0.96**, indicating strong overall detection reliability and balanced classification capability.

The high F1-score demonstrates the effectiveness of the proposed human-centric wearable number recognition pipeline in minimizing both false-positive and false-negative detections.

6.5 Confusion Matrix Analysis

Figures 6.6 and 6.7 present the confusion matrix and normalized confusion matrix for the wearable number recognition model. The matrices demonstrate strong classification accuracy across multiple numerical classes with minimal inter-class misclassification.

Most wearable number classes achieved high prediction accuracy, while only minor confusion was observed among visually similar numerical patterns under low-resolution surveillance conditions. The normalized confusion matrix further confirms stable class-wise recognition capability and balanced detection performance across the dataset.

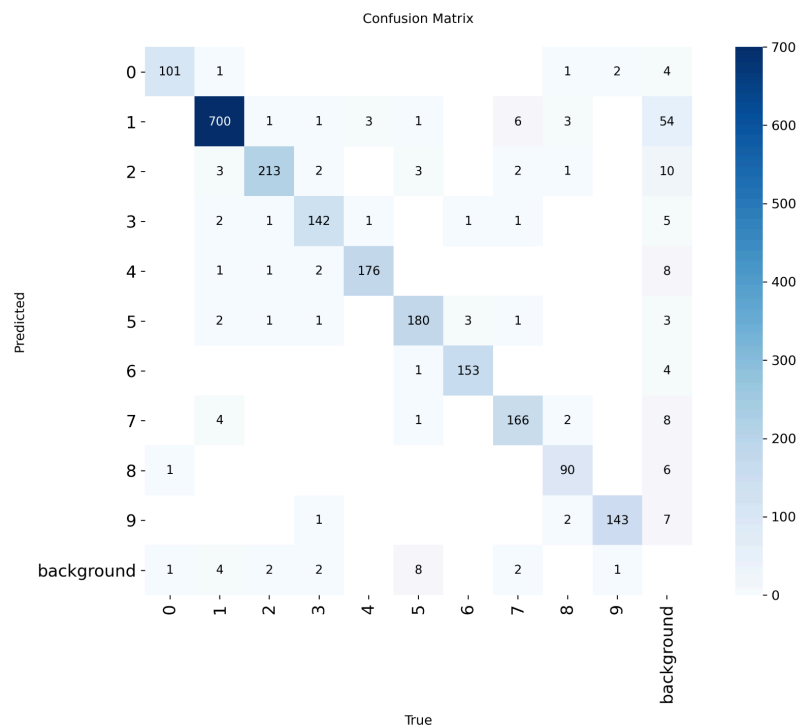


Figure 6.6: Confusion Matrix for Wearable Number Recognition



Figure 6.7: Normalized Confusion Matrix for Wearable Number Recognition

The proposed DetectWear framework successfully integrates YOLOv8n-based human detection with YOLO11n-based wearable number recognition for real-time industrial surveillance applications. Experimental evaluation demonstrated that the system was capable of accurately detecting workers, identifying wearable numerical identifiers, and generating automated alerts for non-compliant personnel within CCTV streams.

The human-centric detection strategy significantly reduced false detections from environmental numerical patterns such as machinery labels, warning signs, and industrial text. In addition, the integration of automated alert generation, evidence logging, GUI visualization, MP4 output generation, and CSV-based alert export established a complete intelligent workforce monitoring and compliance verification framework suitable for secure industrial environments.



Figure 6.8: Sample Wearable Number Detection Results using YOLO11n

Figure 6.8 illustrates sample output results obtained from the proposed YOLO11n-based wearable number recognition model. The model successfully detected and localized visible wearable numerical identifiers from multiple players under varying pose, illumination, scale, and motion conditions. Bounding boxes were accurately generated around the detected numerical regions, while the corresponding wearable identification numbers were classified correctly in most surveillance scenarios.

The experimental results demonstrate the capability of the proposed deep learning-based framework to recognize wearable numerical identifiers even under low-resolution and partially occluded conditions commonly observed in CCTV footage. The model additionally maintained stable detection performance across varying player orientations, background complexity, and movement patterns.

The human-centric recognition strategy significantly reduced false detections from unrelated background numerical patterns by restricting wearable number detection only to detected human regions. Overall, the obtained results validate the effectiveness of the proposed YOLO11n wearable number recognition framework for real-time industrial surveillance and workforce monitoring applications.

6.6 APP UI FOR THE PROJECT

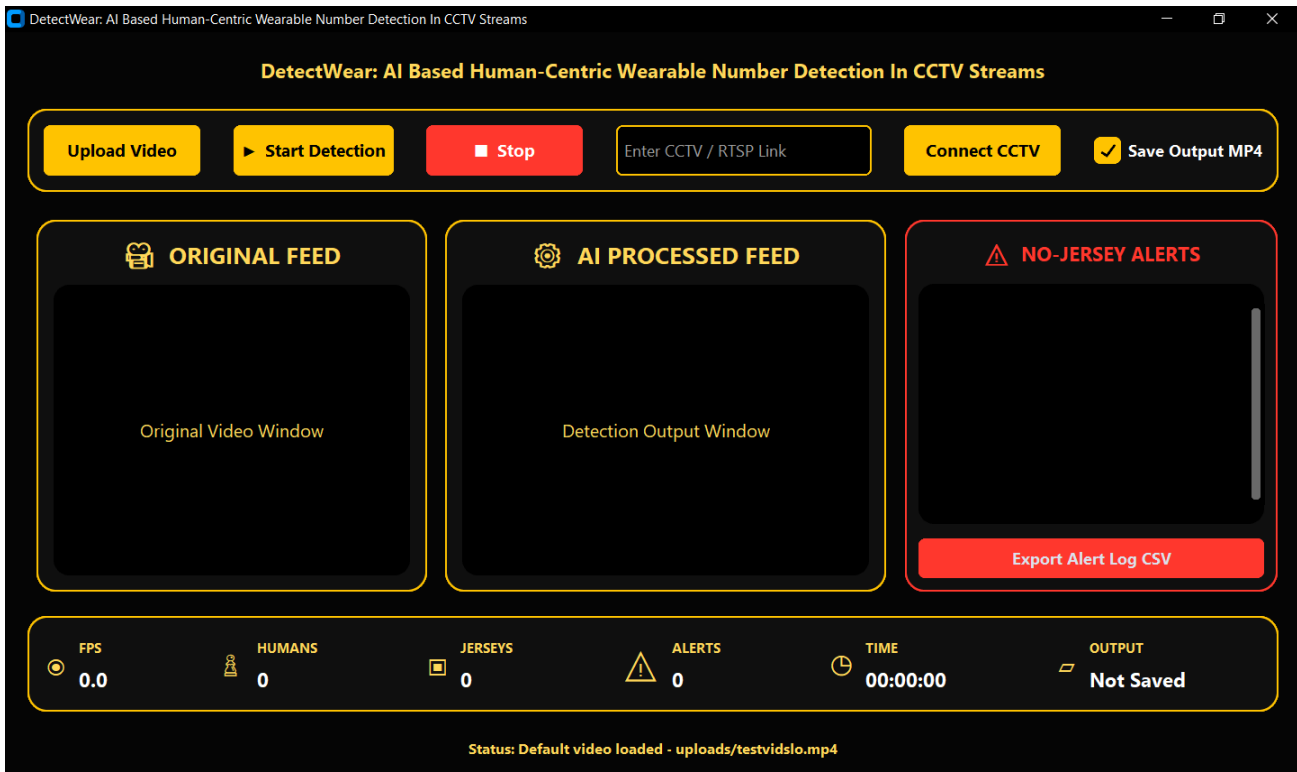


Figure 6.9: Tkinter-Based GUI for AI Based Human-Centric Wearable Number Detection In CCTV Streams

Figure 6.9 illustrates the developed DetectWear graphical user interface designed for real-time industrial surveillance and workforce compliance monitoring. The GUI was implemented using the CustomTkinter framework and integrates video upload functionality, live CCTV/RTSP stream support, wearable number detection visualization, alert monitoring, runtime analytics, and output management within a unified dashboard.

The interface simultaneously displays the original surveillance feed and AI-processed detection feed, enabling real-time comparison between raw and annotated outputs. Operational statistics including FPS, human count, wearable number count, alert count, runtime duration, and output status are continuously updated during execution. The system additionally incorporates a dedicated no-jersey alert panel for displaying compliance violations along with CSV-based alert export functionality.

The developed GUI significantly improves usability, operational monitoring, and real-time surveillance management, thereby enhancing the practical deployment capability of the proposed DetectWear framework within secure industrial environments.

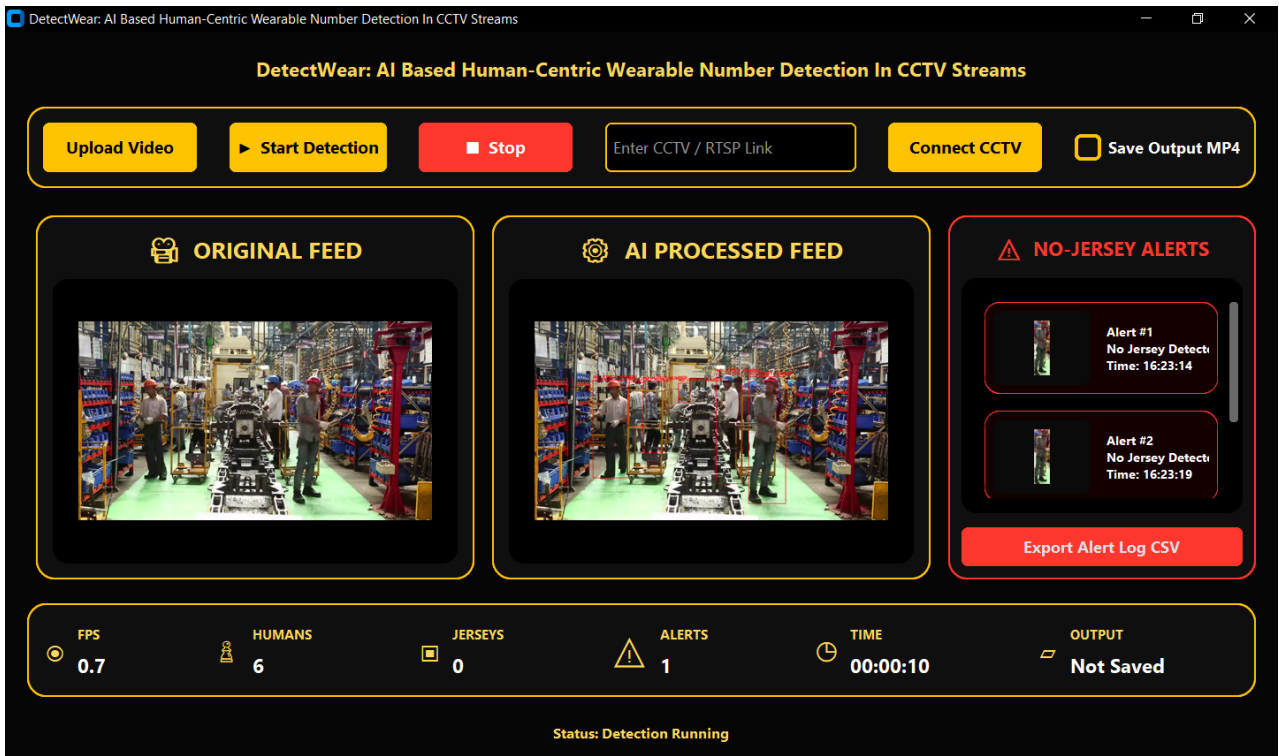


Figure 6.10: Real-Time Industrial Surveillance Monitoring and No-Jersey Alert Generation using DetectWear

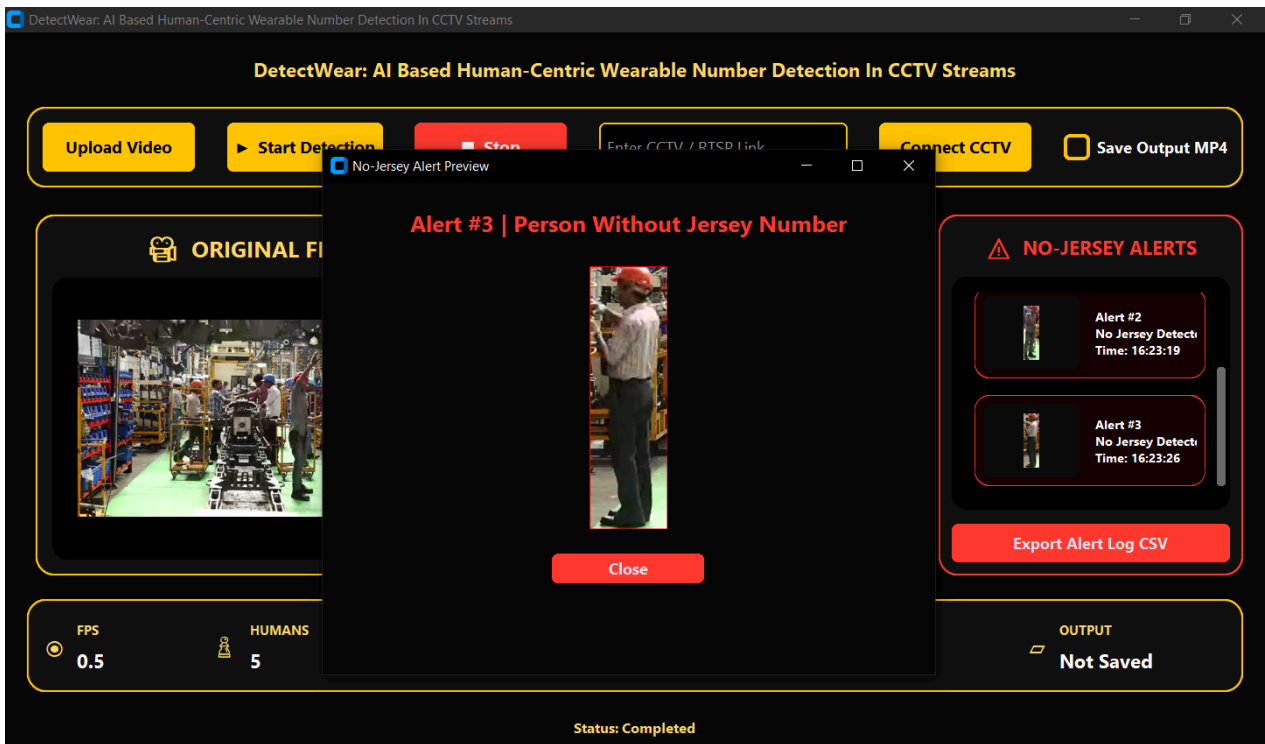


Figure 6.11:Pop-UpAlert Window for Real-Time Detection of Personnel Without Wearable Identification Number

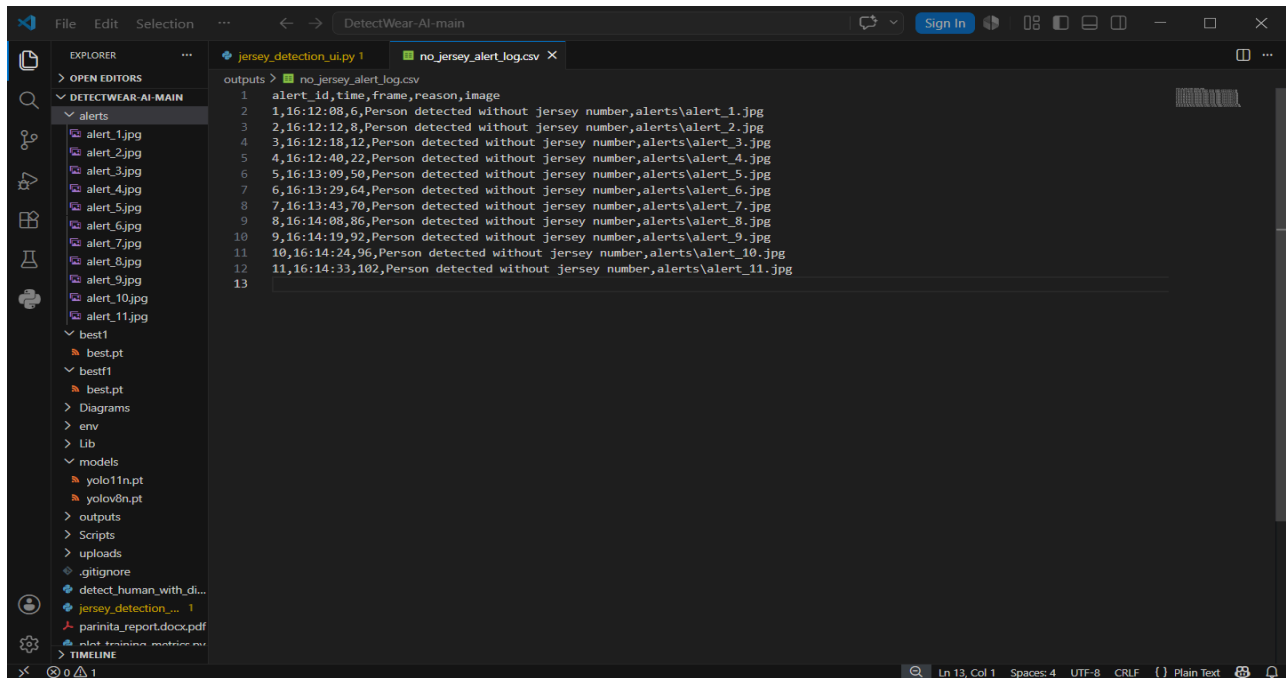


Figure 6.12: CSV-Based Alert Report Generation and Evidence Management System

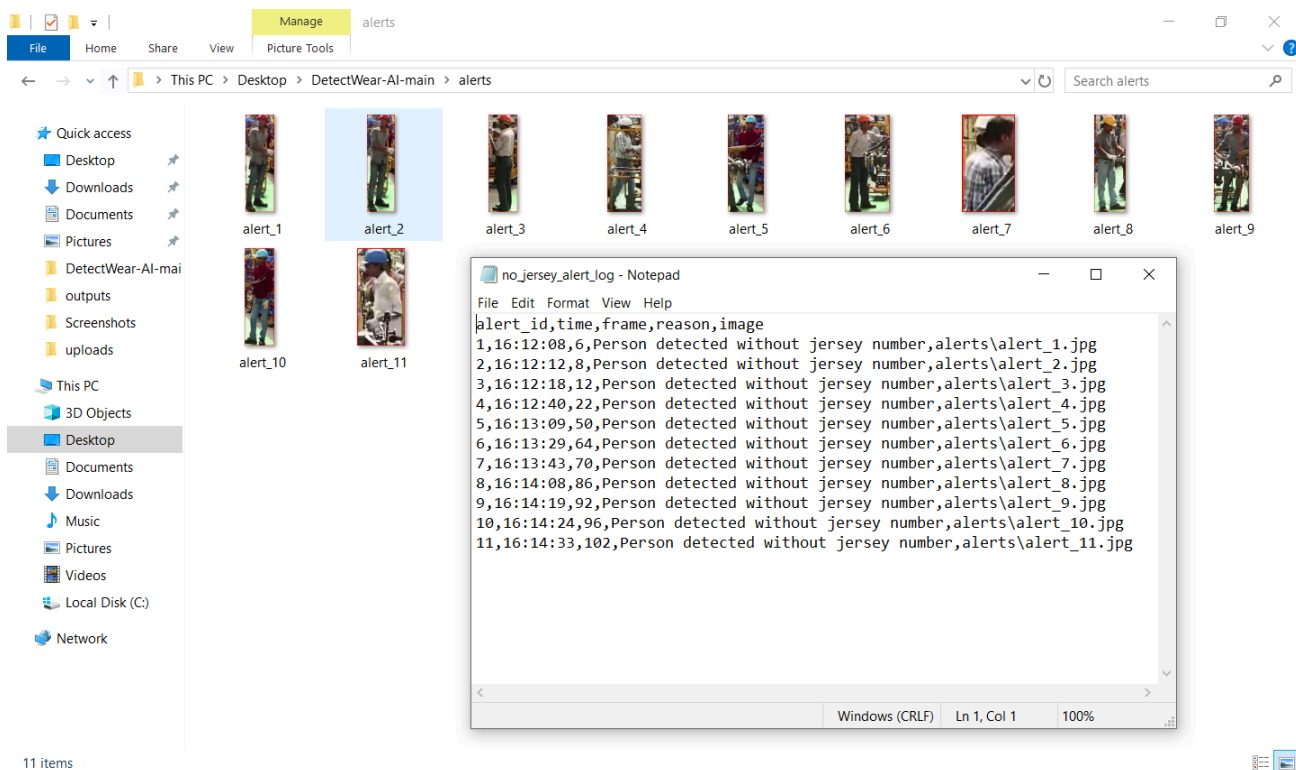


Figure 6.13: Automated Evidence Image Capture and Alert Logging for Non-Compliant Personnel



Figure 6.14: Real-Time Wearable Identification Number Detection and Human-Centric CCTV Surveillance Interface

Chapter 7

CONCLUSION

The proposed DetectWear framework successfully demonstrates the feasibility of integrating deep learning-based human detection, wearable identification number recognition, automated compliance monitoring, and intelligent CCTV analytics within a unified industrial surveillance system. The framework was specifically developed for secure industrial environments such as SDSC SHAR and Vehicle Assembly Buildings (VABs), where real-time workforce monitoring and personnel compliance verification are critical for operational safety and security.

The system employs YOLOv8n for efficient real-time human detection and YOLO11n for wearable identification number recognition within detected human regions. By adopting a human-centric inference strategy, the framework effectively reduces false-positive detections from machinery labels, warning signs, environmental text, and unrelated numerical patterns commonly present in industrial environments. The wearable number recognition model trained using the Roboflow dataset demonstrated strong detection performance with stable precision, recall, F1-score, and mAP metrics under varying CCTV surveillance conditions.

The developed framework additionally incorporates automated no-number alert generation, evidence image capture, CSV-based alert logging, MP4 output generation, and a real-time surveillance dashboard for operational monitoring. Experimental evaluation confirmed that the system is capable of accurately detecting workers, recognizing wearable identification numbers, and generating automated compliance alerts under practical industrial conditions involving illumination variation, worker movement, partial occlusion, and multi-person scenarios.

Overall, the proposed DetectWear framework establishes a scalable, efficient, and practical AI-powered surveillance solution suitable for secure industrial facilities requiring continuous workforce monitoring, restricted-area personnel verification, and intelligent CCTV-based compliance analytics. The framework can significantly assist surveillance operators in improving workforce traceability, operational security, and automated personnel management within mission-critical environments such as SDSC SHAR.

Chapter 8

FUTURE SCOPE

Although the proposed DetectWear framework successfully demonstrates real-time human detection, wearable identification number recognition, and automated compliance monitoring within industrial surveillance environments, several enhancements can be incorporated in future developments to improve scalability, intelligence, and operational capability.

Future work may include the integration of multi-camera synchronization and persistent worker tracking for monitoring personnel movement across multiple surveillance zones within large industrial facilities such as SDSC SHAR. Advanced identity association techniques can additionally be incorporated to maintain continuous worker tracking even under partial occlusion and viewpoint variation.

The framework can further be extended using temporal activity analysis and behavior monitoring techniques for detecting unauthorized activities, restricted-area violations, and abnormal workforce movement patterns. Integration with centralized surveillance databases and attendance management systems may also improve administrative workforce monitoring and operational analytics.

Future enhancements may additionally focus on improving wearable number recognition under extremely low-resolution CCTV conditions through transformer-based architectures, super-resolution techniques, and advanced video enhancement models. The incorporation of edge-device optimization and distributed inference mechanisms can further improve deployment flexibility for large-scale industrial surveillance infrastructures.

Furthermore, the proposed framework can be extended to support additional industrial safety compliance tasks such as PPE verification, helmet detection, safety vest monitoring, and restricted-zone access control within a unified intelligent surveillance ecosystem.

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APPENDIX

Section No.	Description	Pg. No.
1.1	Table 1.1: Detailed Literature Review	11
2.2	Table 2.1: Week 1 Workflow	13
2.3	Table 2.2: Week 2 Workflow	14
2.4	Table 2.3: Week 3 Workflow	14
2.5	Table 2.4: Week 4 Workflow	15
2.6	Table 2.5: Week 5 Workflow	16
4.1	Table 4.1: Comparison of YOLOv8n Model Variants Based on Performance and Complexity	20
4.2	Table 4.2: Comparison of Number Detection Methods	21
4.3	Table 4.3: Comparative Analysis of Surveillance Approaches	22
5	Figure 5.1: Overall System Architecture	23
5.2	Figure 5.2: Dataset Preparation and Augmentation Pipeline	25
5.3	Figure 5.3: Real-Time Video Processing Pipeline	26
5.4	Figure 5.4: Alert Generation Pipeline	27
5.7	Figure 5.5: Snippet of jersey_detection_ui.py code	29
5.7	Figure 5.6: Snippet of train_yolo11_digits.py code	30
5.7	Figure 5.7: Snippet of train_yolo11_humans.py code	31
6.1	Figure 6.1: Training and Validation Performance Metrics of YOLO11n Wearable Number Detection Model	33
6.2	Figure 6.2: Precision-Recall Curve for Wearable Number Recognition	34
6.3	Figure 6.3: Precision-Confidence Curve	35
6.3	Figure 6.4: Recall-Confidence Curve	35
6.4	Figure 6.5: F1-Confidence Curve for Wearable Number Recognition	36
6.5	Figure 6.6: Confusion Matrix for Wearable Number Recognition	37
6.5	Figure 6.7: Normalized Confusion Matrix for Wearable Number Recognition	37

6.5	Figure 6.8: Sample Wearable Number Detection Results using YOLO11n	38
6.6	Figure 6.9: Tkinter-Based GUI for AI Based Human-Centric Wearable Number Detection In CCTV Streams	39
6.6	Figure 6.10: Real-Time Industrial Surveillance Monitoring and No-Jersey Alert Generation using DetectWear	40
6.6	Figure 6.11: Pop-Up Alert Window for Real-Time Detection of Personnel Without Wearable Identification Number	40
6.6	Figure 6.12: CSV-Based Alert Report Generation and Evidence Management System	41
6.6	Figure 6.13: Automated Evidence Image Capture and Alert Logging for Non-Compliant Personnel	41
6.6	Figure 6.14: Real-Time Wearable Identification Number Detection and Human-Centric CCTV Surveillance Interface	42